Public Cloud Deployment Methodology

Professional Services
PSO Methodology

Prepared for: *PAN-PSO*

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Professional Services
Public Cloud Team

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1. Overview

This document is intended to be a comprehensive Professional Services methodology guide on delivering Quick Starts (QS) and Time and Materials (T&M) engagements in Public Cloud environments. Professional Services is tasked with deploying virtual Palo Alto Next Generation Firewalls (NGFW) to public cloud providers including Amazon (AWS), Azure, Google Cloud (GCP), Oracle (OCI), IBM, and Alicloud. As such, it is also important to understand tunneling between On-Premise networks and the cloud, deploying Global Protect on virtual PAN devices, and Panorama deployments in the public cloud.

PS Consultants are expected to deliver projects within the scope of the Service Descriptions listed on the company website. Program Managers (PMs) will seek consultants with time on their calendars in the **#ps-nam-schedule-vm-series** Slack Channel. Technical questions can be addressed promptly from the Global Solutions Architects and other PS Engineers in the **#pso-software-ngfw** Slack channel.

This document covers the following topics:

- Describes the PS Cloud project lifecycle
- Training to prepare consultants for public cloud project delivery
- Internal Lab access for customer project testing.

While Professional Service Org may be tasked with deployment and configuration of the following, the following items in this list are beyond the scope of this document:

- 5g and mobile technologies
- Global Protect
- Private Cloud virtualizations (OpenStack, OpenShift, VMware NSX/VSX, etc.)
- CN series containerized virtual firewalls
- Kubernetes
- Automation work such as Python and Ansible.
- XML and RESTful API automation work, configuration commands, automation of batch configuration updates on virtual NGFW and Panorama.
- Cover every possible Public Cloud Deployment use-cases.

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Additional information regarding Professional Services deployment work can be found in the PSC Runbook. Consultants are strongly encouraged to read and understand the contents of that document.

2. PS Cloud Engagement Lifecycle

Public Cloud Quick Start and (T&M) deployment engagements tend to follow the same general workflow. The typical goal is to deploy and configure virtual NGFW, log collectors, and virtual Panorama management devices. It can be helpful to visualize deployment activities as distinct phases as seen in figure 1.

Design	Deploy	Configure	Document
Work with the customer to understand their traffic flows. Align their needs with our recommended arch and best practice.	Get the firewalls up and running in the customer environment.	Work with customer to understand their traffic. Align their needs with our arch and best practice.	Work with customer to understand their traffic. Align their needs with our arch and best practice.
Kickoff **Internal and external** Verify licensing, deployment profiles **TRD Session** **Capture Customer needs** Create network diagrams, planning Triggers Terraform creation, test Terraform In our labs	**Execute Terraform** **Infra as Code** **Initial FW Config** **Immediate tasks to make the NGFW usable** Set the admin pass Connect to Panorama	**Config Tasks** **The "Day One" tasks** Templates & Device groups, Interfaces, Health Checks Load Balancers Routing Backup new Configs	**As Built** **The "final" deliverable** Request TSF files from Customer Run the SwiftDoc tool **Cleanup** **Final Tasks** Finalize the TRD Add all documents and images to the project repo Push the changes back to Spring

Figure 1 - Cloud Project - Typical Project Lifecycle and Milestones

Each silo has tasks that should be completed. These are considered a gating mechanism to the next phase of the project.

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For the design to be considered complete, the consultant should work with the customer to populate a TRD template that captures the customer design including current and planned network diagrams.

For deployment to be considered complete, the consultant should have the firewalls up and running in the customer public cloud, including licenses and Panorama connectivity.

2.1. Collection and Handling of Customer Data

Professional Services Consultants are required to comply with Palo Alto Networks Information Security Policy.

All customer data must be labeled as directed in the company Information Classification and Handling Standard. According to the standard, customer data should be treated as "restricted". Be sure to transfer and store customer data as outlined in the Customer File Sharing section.

- All Customer information collected via Palo Alto Networks Products and Services and deemed by the customer as sensitive, confidential, or restricted;
- Customer authentication credentials.

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Information Labeling

Classification	Requirements
	"Palo Alto Networks Secret" must appear on each page. Control statements, as shown in note "c" below, which clarify a label, must be added directly under the "Secret" label on each page. "Secret" must appear on removable media labels. Filename must not indicate the classification or sensitivity of the information.
Restricted	"Palo Alto Networks Restricted" must appear on each page. Control statements, as shown in note "c" below, which clarify a label, must be added directly under the "Restricted" label on each page. "Restricted" must appear on removable media labels. Filename must not indicate the classification or sensitivity of the information.
Confidential	"Palo Alto Networks Confidential" must appear on each page. Control statements, as shown in note "c" below, which clarify a label, must be added directly under the "Confidential" label on each page. "Confidential" must appear on removable media labels.
Public	No labeling is required.

2.2. Services

Beyond the "standard" QS and T&M engagements, Professional Services does offer other types of engagements that cloud-focused consultants may be asked to perform.

2.2.1. Design and Deployment Validation

There is a pair of SKUs for design and deployment validation, as described at this link, VM-Series for Public Cloud Design Validation and Deployment Validation

2.2.2. Global Protect

PS does deliver Global Protect as part of Strata VM series deployments. Refer to the GlobalProtect-PSO-Methodology for technical details on deployment of GP.

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2.2.3. Terraform Add-On Service

During a Quick Start engagement, the Terraform that is used for deployment is and remains the property of Palo Alto Networks. We must provide deployment code to the customers for deployment purposes. If the customer wants to modify or keep that code, for integration into their CI pipeline for example, it may be done so at the discretion of the PS engineer. The customer should be made aware that they are taking the ownership of the Terraform, without official support. Professional Services offers a service to assist customers with the automation of infrastructure as code deployments using Terraform. This Terraform add-on service is described at this link.

3. Training

A Public Cloud provider like Google, Azure, AWS, etc. allows folks to rent slices of their network. The slice you (the tenant) rent is often referred to as a "Virtual Private Cloud" (VPC). The tenant has administrative control over the resources in their VPC. Creating virtual networks and servers in your VPC is what people mean by hosting them "in the cloud".

Enablement is learning for the customer. Training is general learning like how the firewall works, how to do generic tasks, etc.

3.1. Required Training

Ideally a PSC will complete self-directed training on the fundamentals. Once this is complete, they would join the instructor-led training before they begin shadowing. Of course this may not be possible if the instructor led training is not available in a timeframe that fits the schedule of the PSC, conflicts with customer work, etc.

This is a list of the self-directed training that each PSC is expected to complete in preparation for instructor-led training:

- Pre-requisite training and fundamentals are available at the following link. This self paced training needs to be completed before attending the in-person workshop.
- After completing the fundamental training, submit to attend an upcoming instructor-led workshop. These are now being administered by providers (AWS, GCP, Azure only).

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- Additional NGFW training information is described in this slide deck.
- The PS Training / Professional Services Academy page describes additional training and documentation that may be helpful.
- Palo Alto Networks Micro-Credential for Remote Security Consultant (PMRC)

3.2. Shadowing/Reverse Shadowing

Shadowing means the PSC finds a senior consultant to mentor them by joining customer facing meetings to get a better understanding of how a project works. Work with your line manager to find shadowing opportunities.

For reverse shadowing, the senior PSC doing the reverse (lead) role should book the billable time in Clarizen. The person that is shadowing logs their time in Clarizen as enablement or training.

3.3. Requisite Skills (General)

These general skills are useful for the PSC to develop.

- Bash Shell Scripting
- Git, Github, and Spring
- Terraform TF Enablement Training
- Familiarity with Python3.

3.4. Requisite Skills (Cloud Infra)

Consultants should develop proficiency in the following topics in preparation for public cloud engagements.

- Public Cloud - AWS
- Public Cloud - Azure
- Public Cloud - GCP
- Public Cloud - IBM
- Public Cloud - OCI
- Infrastructure as Code (Terraform)

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- Tunnels (IPSec, VPN, etc)
- Load Balancers
- HIgh Availability (HA) Configuration

4. Tools

This section describes how to obtain and configure various tools.

4.1. VS Code

Visual Studio Code is an open source lightweight source code editor with rich ecosystems of API extensions. VS Code can be downloaded at the link below:
VS Code Download

4.2. Windows Subsystem for Linux (WSL)

If you are using Windows, you can install WSL. This will give you a BASH shell and an environment similar to what we are using for Linux/Mac.
https://learn.microsoft.com/en-us/windows/wsl/install

4.3. SSH

You will also need to create an SSH key using a method similar to the following:
https://medium.com/devops-with-valentine/2021-how-to-set-up-your-ssh-key-for-github-on-windows-10-afe6e729a3c0

https://docs.github.com/en/authentication/connecting-to-github-with-ssh/generating-a-new-ssh-key-and-adding-it-to-the-ssh-agent?platform=linux
Once your SSH key pair is created, you must add the PUBLIC key to your Github account.
https://docs.github.com/en/authentication/connecting-to-github-with-ssh/adding-a-new-ssh-key-to-your-github-account

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4.4. Terraform Setup

Terraform is a tool created by a company called Hashicorp. Terraform files written in HCL are what you use to define and implement your Infrastructure as Code in your VPC.

Why to Deploy a Firewall in the Public Cloud with Terraform?

The preferred method for deploying Terraform in the customer environment is to pass them a compressed file containing all the Terraform code. The customer can then upload this file to the cloudshell of the Public Cloud provider, where it can be unpacked. The consultant should review the Terraform code files, especially the terraform.tfvars and variables.tf files to make sure everything is correct before the deployment.

https://docs.paloaltonetworks.com/best-practices/10-1/best-practices-for-managing-firewalls-with-panorama

Each public cloud provider has a web site that allows tenants to provision resources in their own VPC. This is great for smaller projects, but point and click provisioning does not scale well for companies looking for enterprise-grade network solutions and hosting of high-traffic applications in the cloud.

Cloud providers offer an Application Programming Interface (API) as a way to provision resources that is less of a burden on human operators.

As detailed on the Azure web site, Infrastructure as Code (IaC) is the programmatic management of public cloud network infrastructure.

Use the following steps to set up the Terraform binary on a host. Note that the version of Terraform you choose should be matched with the code you are attempting to deploy. There may be version constraints that are usually found in the `providers.tf` or `versions.tf` files.

4.4.1. Azure Cloud Shell

- Azure cloud shell instructions (Use BASH, not PowerShell)

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4.4.2. AWS Cloud Shell

In the AWS Cloud Shell, you first install `yum-config-manager` to manage your repositories.

```
Unset
sudo yum install -y yum-utils
```

Next, use `yum-config-manager` to add the official HashiCorp Linux repository.

```
Unset
sudo yum-config-manager --add-repo
https://rpm.releases.hashicorp.com/AmazonLinux/hashicorp.repo
```

Finally, you can install Terraform as usual from the Linux command line.

```
Unset
sudo yum install -y terraform
```

4.4.3. Google Cloud Shell

Terraform should already be installed in GCP Cloud Shell.

- Google Cloud shell instructions

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4.4.4. Local Workstation

Installing Terraform on your local machine is useful for testing purposes. It is good to try out the customer code and verify the `tfvars` file before it is time for the actual deployment.

Go to the Hashicorp website by following this link. Select your operating system and follow the directions you find there to install Terraform.

4.5. CLI Tools

Each public cloud provider has a CLI tool that can be used on your local workstation to help manage your deployments and remote infrastructure configuration. These tools are often pre-installed in the "cloud shell" of each provider as well. It is good to install these tools and become familiar with them since you may need to assist our customers with them.

This table lists each Cloud Provider and the link to the respective provider site:

Provider	**CLI Tool**
AWS	AWS CLI
Azure	Azure CLI
GCP	GCP CLI
OCI	OCI CLI
IBM Cloud	IBM Cloud CLI
Alibaba Cloud	Alibaba Cloud CLI
OpenShift	OpenShift CLI

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4.6. Brew for Apple Mac

You have to install developer tools on Mac to use Brew. The reason is, Apple has a proprietary compiler and tool chain that they require you to install.

From a terminal window:

```
Unset
xcode-select --install
```

Now you can install Brew by following the directions. https://docs.brew.sh/Installation

4.7. Lucid Chart

The web based Lucid Chart application can be used to create network and traffic flow diagrams. Use Sheldon bot in Slack to request a LucidChart license by asking "how can I get a LucidChart license" and following the prompts. The LucidChart web based application.

There are many shared examples of these types of diagrams that you can use as starting points in your projects.

https://lucid.app/documents#/documents?folder_id=180215948

4.7.1. Creating a Diagram in Lucid Chart

- Navigate to the Okta dashboard and start the LucidChart application.
- Create a new folder if needed, then a new blank document.
- Under the "view" tab, uncheck the "Infinite Canvas" box.
- Under "file" and "document settings", change the orientation to "landscape".
- Change the paper size to "A5" in doc settings.

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5. Lab and Testing Accounts

Use the following steps to request internal lab access:

- Open an IT ticket to join "PSO-CLOUD DL" distribution list with the Enterprise Service Portal
- Click on "Request Something"
- Select "Service Request"
- Choose Type "Email / Outlook / Distribution List"
- Choose Category "Add user to email alias"

Refer to PS Cloud Lab Accounts for more information about getting lab access and proper procedures while using the labs. You may also find some helpful information here: TAC/PS Cloud Labs Access and Policies

If you will be using tools that require API access (terraform, aws-cli, boto3, etc), some additional setup will need to be done. See the section Terraform Setup for specific details about configuring Terraform. See the section Terraform Execution for details about using Terraform to deploy customer infrastructure.

5.1. Internal Lab: Alicloud

5.2. Internal Lab: Amazon Web Services

Access to the AWS Lab account is controlled by Okta Universal IDP. Okta SSO for AWS Universal IDP utilizes the Okta App portal to provide SSO access to AWS. The user can choose a specific role when connecting. Your role determines what permissions you have, using AWToolsS IAM permissions attached to the selected role. AWS Universal IDP is the current security standard for all AWS accounts company-wide.

Full details about access to PANW AWS accounts can be found at Information Security Confluence

To log into the environments,

- Authenticate to Okta Web Portal.

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- Select on the AWS Universal IDP application and complete MFA.

For the PS Lab Account, select:

- For account **pso-shared-lab-fy23q1** (**751512424814**)
- Select **Role sso_engineer**

If you have access to any other AWS accounts or roles, they will also be listed here. There are two AWS accounts that will be deactivated on **Monday, April 3rd 2023**. Note, these are the same accounts that were previously communicated but the deactivation date was delayed.

Account Name	Account ID	Status
IT_AWS-PS-Architects	632512868473	Decommissioning April 2023
pso-shared-830100	354128141335	Decommissioning April 2023

5.2.1. Okta AWS CLI

Use the `okta-aws-cli` utility to obtain AWS credentials on your local machine if you need them for Terraform. The `okta-aws-cli` is native to the Okta Identity Engine and its authentication flows. The Okta AWS Federation application is SAML based and the Okta AWS CLI interacts with AWS IAM using AssumeRoleWithSAML. `okta-aws-cli` handles authentication through Okta and token exchange with AWS STS to collect a proper IAM role for the AWS CLI operator. The resulting output is a set made up of Access Key ID, Secret Access Key, and Session Token of AWS credentials for the AWS CLI. The Okta AWS CLI expresses the AWS credentials as either environment variables or appended to an AWS CLI credentials file.

NOTE: The Session Token has an expiry of 60 minutes.

Use these steps for getting `okta-aws-cli` to work with your AWS lab account.

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5.2.2. Steps to Install aws-okta-cli

Macintosh:

- brew install okta-aws-cli

Linux/Windows:

- Download binaries from here

5.2.3. Configure aws-okta-cl

Copy this text into your `~/.env` file.

```
Unset
OKTA_ORG_DOMAIN=paloaltonetworks.okta.com
OKTA_OIDC_CLIENT_ID=0oaripjsjwzdfRyCm1t7
OKTA_AWS_ACCOUNT_FEDERATION_APP_ID=0oae0k7sqyScYDeY31t7
OKTA_AWSCLI_IAM_IDP=arn:aws:iam::751512424814:saml-provider/OKTA_SSO
OKTA_AWSCLI_IAM_ROLE=arn:aws:iam::751512424814:role/sso_engineer
OKTA_AWSCLI_EXPIRY_AWS_VARIABLES=true
OKTA_AWSCLI_SESSION_DURATION=28800
OKTA_AWSCLI_OPEN_BROWSER=true
OKTA_AWSCLI_CACHE_ACCESS_TOKEN=true
```

- Run the below command

```
Unset
okta-aws-cli -o paloaltonetworks.okta.com -a 0oae0k7sqyScYDeY31t7
--oidc-client-id 0oaripjsjwzdfRyCm1t7 -i
arn:aws:iam::751512424814:saml-provider/OKTA_SSO -z
```


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Above command will open the URL in a new tab, please follow the steps to begin Okta device authorization for the AWS CLI. This command will create AWS credentials and stored in `~/.aws/credentials`

- Finally, verify AWS CLI is working:

```
aws s3 ls
```

5.3. Internal Lab: Azure

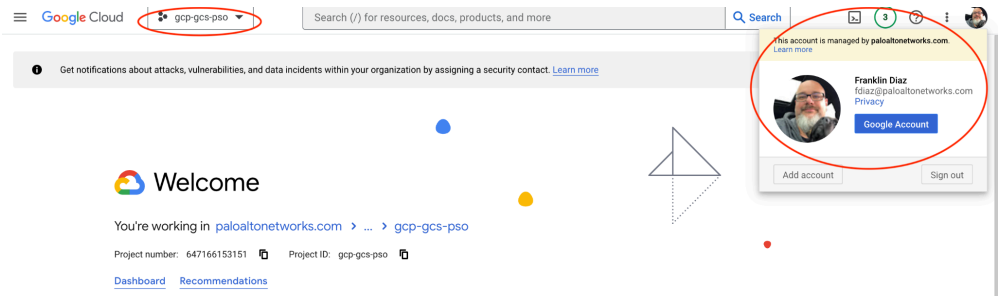
To access Azure Cloud resources, log in at the following link: [Microsoft Azure Portal](https://theloop.paloaltonetworks.com/loop/public-cloud-lab-azure)

https://theloop.paloaltonetworks.com/loop/public-cloud-lab-azure

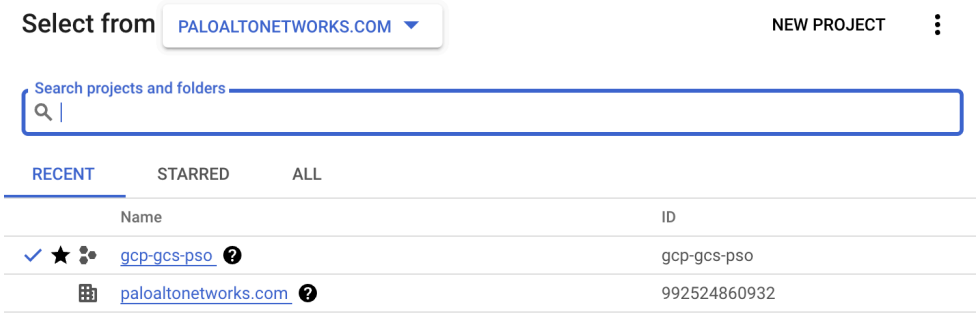
Check the following link if you are having difficulty connecting your workstation to the Public Cloud provider: Using Global Protect when accessing Azure and GCP

5.4. Internal Lab: Google Cloud

To access the GCP lab, use your browser to get to https://console.cloud.google.com. Be sure you are logged in with your Palo Alto Networks account by checking in the upper right.



You also need to verify you are in the “PALOALTONETWORKS.COM” with project name “gcp-gcs-pso”.



Once your credentials are working, you may wish to add a dedicated "Terraform" user.

The "gcloud" CLI tool is also a useful tool in lab work and deployments. Use the gcloud CLI overview to learn more about the tool, including how to install it on your local workstation.

Check the following link if you are having difficulty connecting your workstation to the Public Cloud provider: Using Global Protect when accessing Azure and GCP

5.5. Internal Lab: IBM Cloud

- Training slide deck on IBM cloud VM series
- video: VM series on IBM cloud

5.6. Internal Lab: Oracle OCI

Send a request to helpdesk@paloaltonetworks.com to get added to OCI. For now, the only way to have a working account is to request for "Federated user account with OracleIdentifyService with admin access to OCI."

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Here is an excellent walk through on how to get lab access to OCI and deploy VM series firewalls: VM-Series on Oracle Cloud High Availability deployment guide_Mar-2023

Follow these links for internal documentation on working with NGFW VM series in Oracle Cloud:

- https://theloop.paloaltonetworks.com/loop/oci
- https://drive.google.com/drive/folders/1qHLhs-mr2zDf7l7vPi9xzzRM69Vm7E7s
- https://docs.oracle.com/en/solutions/secure-app-palo-alto-firewall/index.html

To access Oracle Cloud lab internally:

- Navigate / go to https://console.us-ashburn-1.oraclecloud.com/
- In the Cloud Tenant box, type in “panwgcs”.
- Under the “identity provider” drop down, select “Okta” and click “continue”.
- Sign in using Okta.

5.7. Serial Numbers and Auth Codes

Log in to the Customer Support Portal to create a deployment profile for your lab work. It is important to remove old/unused profiles after you test to release the credits back to the pool.

There is a new process for how PSC will request and use FLEX credits in the lab as detailed at this link:
https://spring.paloaltonetworks.com/pso-prisma-automation/ps_documentation/blob/main/02-Lab/serial_numbers.md

5.8. Palo Alto Networks External Github

Palo Alto Networks maintains a company branded Github account that is available to the public. It may not be relevant to most customer cloud

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deployments, but the information is included here because we do have several "Terraform Module" repositories under active development. The Terraform modules we use in our deployments and make available to the public can be found at this link: https://registry.terraform.io/namespaces/PaloAltoNetworks

Use the following links to get access and learn more about Github:

- https://github.com/PaloAltoNetworks/.github/blob/master/Learn-GitHub.md
- https://theloop.paloaltonetworks.com/loop/github-at-palo-alto-networks

If there are any issues getting access or using Github, you can reach out to #github-questions on Slack.

5.9. Spring Enterprise - Internal Github

Spring is an internal private version of Github Enterprise. There are more details about Spring here on Spring: https://spring.paloaltonetworks.com/btorres-gil/spring

Our shared code repository on Spring is located at this link: https://spring.paloaltonetworks.com/pso-prisma-automation

You must be added as a member to the "pso-public-cloud" group on this repository before you will be able to see all the code stored at this location. You must also authenticate by clicking the "Spring" tile in Okta.

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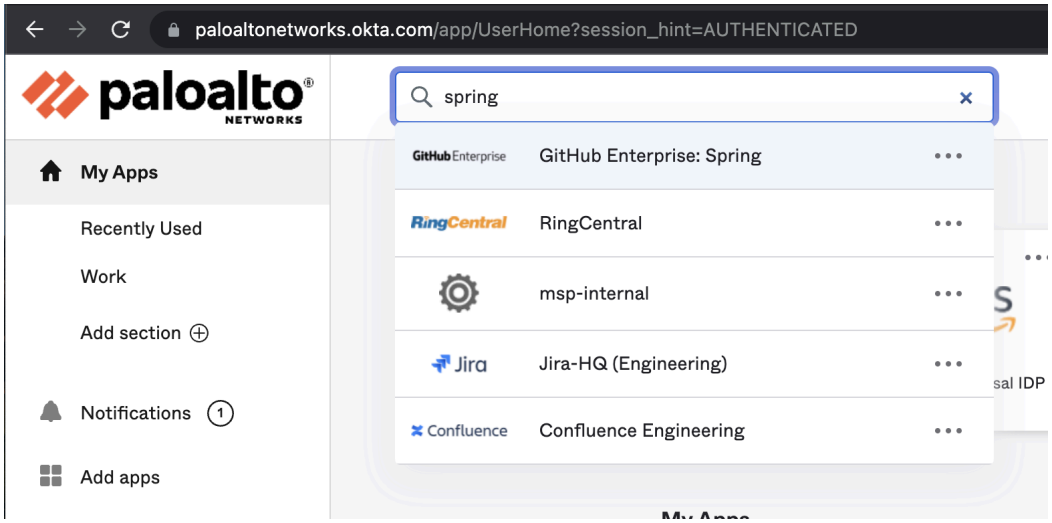
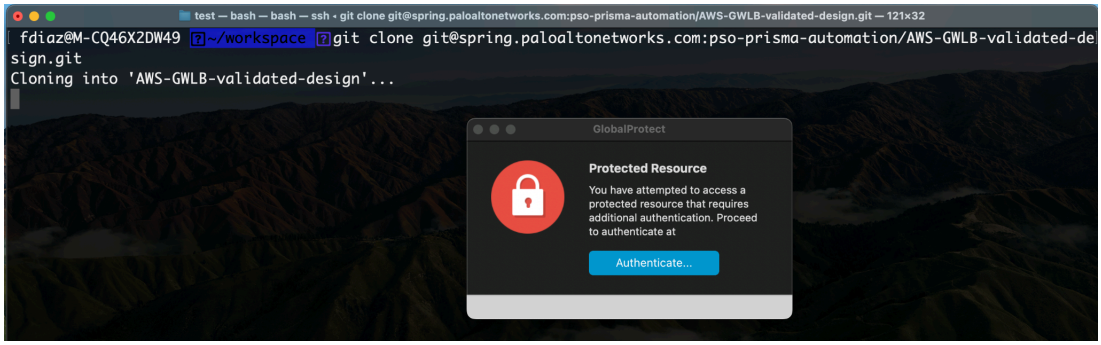


Note that you must authenticate separately/again for command line access to Spring repositories as seen here:



5.10. Salesforce

Before you can access Salesforce, you must request access.

- Go to Service Now - ServiceNow
- Click on IT Services => Request Something => Services => Service Request
- On the Form:
  - Type = Salesforce
  - Category = Access/Permission
  - Subcategory = Grant access to new user
- Fill out a short description

- In the long description, indicate your role. To make it simple, ask an existing user in your team and ask them to copy the same profile/role as that individual.

Once you have successfully requested access, use the "SFDC" tile in Okta for Salesforce access.

For general Salesforce questions, send an email to GCS-Ops-Access
There is also a Slack channel #knowledgebase

5.11. Lab Panorama

An existing HA pair of licensed Panoramas is running in GCP. It is encouraged to use the shared lab Panorama in an effort to reduce lab costs. You can connect your workstation and customer/lab firewalls to these for testing purposes.

The username is "admin" and the IP addresses are as follows:

- 34.134.31.136 (primary)
- 34.136.90.64 (secondary)

Adding the IPv4 address of your workstation, as well as the IPv4 address of the firewalls you would like to test to an "ingress rule" in Google Cloud allows your hosts to make network connections to the Panorama. Note that certain ports are required for the Panorama to communicate properly with NGFW, allow SSH and HTTPS traffic, and so on.

franklin-lab-ingress	Ingress firewall rule	Global	1000	Apply to all	IPv4 ranges: 20.246.188.228/32 10.199.0.0/21 34.136.90.64/32 20.246.188.24/32 20.121.107.240/32 34.66.44.164/32 35.185.72.81/32 68.38.137.81/32 20.246.188.237/32 34.134.31.136/32	—	tcp:22, 443, 3978, 8443, 28769 icmp

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6. Planning and Design Phase

Design work is a critical part of most deployments. The amount of design work will vary depending on the type of project.

While design work is out of scope for a Quick Start engagement, you are still expected to assist the customer with their questions. Ask the customer to provide a diagram of the planned network design/changes. It is expected that the customer will purchase an additional SKU for design work if they would like design help during their Quick Start engagement. It may be helpful for all involved if the consultant does an additional network diagram that aligns more closely with Palo Alto documentation such as the Architecture Reference Guide and the Deployment Guide for the relevant Public Cloud.

During a T&M engagement, you may be called up to provide diagrams, answer design questions, and be prepared to discuss Palo Alto best practices with our customers.

During the Design phase of a project the PS Consultant is expected to meet with the customer to understand what data and resources they would like to protect.

6.1. Internal Kick Off Meeting

The first engagement the PSC has with the Palo Alto account team and sales team that set up the engagement is usually the internal kickoff meeting.

6.2. External Kick Off Meeting

The first engagement with the customer is typically the “external kick off” where the PM explains the scope of the project to the customer. You may be asked to schedule a Technical Requirements Document (TRD) session at the end of this kickoff meeting.

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TRD sessions are typically up to two hours in length. During this session, the consultant works with the customer to populate the TRD template.

- AWS Reference Architecture
- Azure Reference Architecture
- GCP Reference Architecture

Use the PAN-OS Software Release Guidance page to help the customer choose the best PAN-OS release for their needs.

Verify the customer licensing strategy (FLEX, BYOL, etc.) and assist the customer with creation of deployment profiles in the Customer Support Portal.

If the customer has chosen to deploy with Terraform, the PSC or program manager can reach out to Once enough of the right information is gathered, we let the Poland Scale team know to start review and Terraform creation.

6.3. Project Documentation Folders

Ask the project manager to provide the location of the public and private folders in Google Drive that are used to store project documents.

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PLEASE READ - Process for Giving/Gaining Access to PS Project Folders has Changed Inbox ×

David Kopca Thu, Oct 19, 8:32 AM
to bcc: me

(Bcc: All PS Consultants, PM Managers, and PS Managers)

PMs and Consultants,

Google has changed the way requests are sent out for sharing folders and files within Gdrives. As a result, PMs no longer receive share requests for project folders from consultants and customers.

Action for PMs:

1. Starting immediately, PMs are asked to **proactively provide project folder access to the consultants, sales, SDL and CS, as appropriate, prior to the internal kickoff meeting.**
2. **During the customer kickoff call, the PM should ask for the list of customer emails requiring access**. The PM should inform the customer that Google Share Requests are not monitored. Additional requests must be made directly to the PM. Remember customers should only have access to the Public subfolder and not the entire project folder.
3. If you need access to a project that is not in your theatre, please contact a PM Manager for assistance.

Action for Consultants:

1. If you don't have access to the project folder, please reach out directly to the PM in order to be granted access.

Note: this only affects PS and not EE projects. Katy Edwards is working on a solution for EEs and customers to access EE projects folders.

Best regards,

David Kopca | Senior Manager, Professional Services Project Management
Mobile: 214-918-1788 | www.paloaltonetworks.com

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6.4. Customer File Sharing

It is often necessary to exchange design and configuration data with our customers in a secure manner. For example, when you ask the customer to collect and upload Tech Support Files from their Panorama and NGFW you want to be sure to get those in a secure manner.

PSC should strongly encourage the customer to use secure file transfer methods at all times. It is not recommended to exchange files using methods other than those described here, for example customer instances of Microsoft Teams, etc.

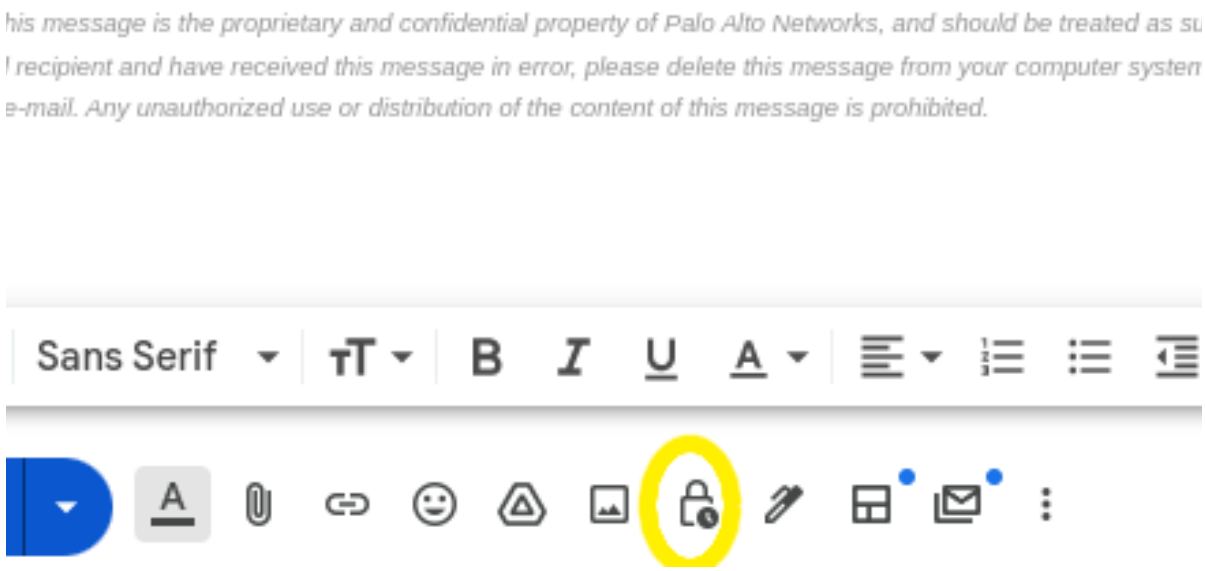
6.4.1. Google Drive

Each project has an associated GDrive folder that can be used to store project artifacts like Terraform code, diagrams, etc. If possible, the customer can up/down load files from here. Be sure to reduce access to the customer data to the least amount of access needed to complete the project.

6.4.2. Email

It is possible to send encrypted email to the customer by adding the “[encrypt]” tag to the beginning of the subject line in your email. This will cause the email to be routed to Proofpoint for encryption. The customer will be required to create a login to view the contents of the email.

Refer to the document [How to: use Proofpoint to send an encrypted email] for more details.



You should also click the padlock icon to toggle “confidential mode” on Gmail. Note the yellow circle in the image.

6.4.3. TAC Cases

You can use a TAC case to exchange files with your customers. See the [Salesforce] section for information on how to get the access you need to view and create cases.

This [Sharing Files with Customers via TAC Case] presentation outlines the steps for using a TAC case for secure file exchange with our customers.

Other resources for managing TAC Support Cases:

- TAC Support Quick reference card for Customers
- Support Resource guide
- Guide to Platinum Support vs. Planned Events
- The Field Escalation Process & Application Training
- When to request a support event

6.5. Network Diagram Creation

6.5.1. Google Cloud Design

- Google Cloud Developer Architecture - GCP diagramming tool
- Google Cloud Developer Cheat Sheet
- 1 - Public Cloud Workshop - GCP Basic - Describes Traffic Flows in GCP

6.6. High Level Design

The spreadsheets seem well-suited to capturing details to be included in HLD but the doc Steve Steiner did looks a lot nicer.
https://docs.google.com/document/d/1XuQsVoE8XXeTl4HDlZ4gVWnubdz3q3qTRLlxFFEPc-k/edit (porto seguro HLD template)

https://drive.google.com/drive/folders/1jiT5ar5dzzpf-ivst6gY_tHKq19R-8aN

Need to add in the “other” HLD templates

6.7. Technical Requirements Document

The Technical Requirements Document (TRD) is used to facilitate a discussion on customer requirements and the expected outcome of the engagement. It is common for a first customer session to populate the TRD to take up to 2

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hours. Follow up discussions may be needed, and often the customer has follow up actions to collect network and design details. The consultant will often be expected to research customer questions and concerns.

During the first meeting with the customer, the PSC should have a copy of the TRD template (see list below) and collect details related to the project. Store the TRD in the

6.7.1. Gather Information on Existing Cloud Infra

Use the cloud scripts to collect JSON data on the cloud deployment in the customer environment. The JSON files contain information about the existing customer cloud environment that you will be asked to integrate with.

https://github.com/devsecfranklin/cloud-scripts

There are some Alpha version processing tools available that can be used to create reports from this data.

6.7.2. TRD Templates

- The Panorama TRD template is available at this link: Panorama TRD Template
- Here is the Terraform add on template: Terraform Add-On TRD Template
- TRD templates and Validation checklists for each provider can be found in the PS Knowledge Base.
- Check the menu on the left side of this page for cloud specific templates. https://theloop.paloaltonetworks.com/loop/professional-services-software-ngfw

6.7.3. Cloud TRD Information

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- Customer to provide current and planned network diagrams for inclusion in the Technical Requirements Document (TRD).
- Customer should verify that they are able to log in to the Support Portal.
- It is assumed that the customer is using Flex licensing. Consultant to verify that this is the case.
- If the customer is using FLEX licensing, the consultant can assist the customer to create a Deployment Profile per the documentation.
- What are the IP addresses of your two DNS and two NTP servers for use in cloud env?
- CIDR blocks for new subnets.

Cloud Provider	Deployment Model	Num of Required Subnets	Recommended Subnet CIDR Size	Minimum CIDR Size per subnet	Maximum Allowed CIDR for VPC/VNet
Google	Shared VPC	3	/24	/29	/8
Azure				/29	/8
AWS	Combined	4		/27	/16
Oracle				/30	/16

Figure 2 - Recommended subnet sizing for supported Public Cloud providers

6.8. Traffic Flow Diagrams

Traffic flow diagrams are created for each flow type (INBOUND, OUTBOUND, EAST-WEST, On Prem) that the customer would like to protect with the security infrastructure we are deploying. The diagrams are created during the design phase so there is a full understanding of the expected outcome of the engagement. These diagrams are also to be included in the "As Built" document as detailed in the Traffic Flow Diagrams section later in this document.

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Some example diagrams are in our shared Lucid charts folder. These can be used as a starting point for your project diagrams.

- LucidChart Projects Folder

6.9. AWS Design

The Gateway Load Balancer (GWLB) was introduced by Amazon in 2022. For more information about GWLB, follow this link:
https://docs.aws.amazon.com/elasticloadbalancing/latest/gateway/introduction.html

The Transit Gateway (TGW) enables you to attach Amazon VPCs, AWS S2S VPN and AWS Direct Connect connections in the same Region, and route traffic between them. Transit Gateways are designed to be highly scalable and resilient. You can attach up to 5000 VPCs to each gateway and each attachment can handle up to 50 Gbits/second of bursty traffic.

https://aws.amazon.com/blogs/architecture/field-notes-working-with-route-tables-in-aws-transit-gateway/

Using overlay routing in your VM-Series firewall integration the AWS GWLB allows you to use two-zone policy to inspect traffic leaving (egressing) your AWS environment. This allows packets to leave the VM-Series firewall through a different interface than that which they entered through.

When overlay routing is configured, the firewall is able to perform a Layer 3 route lookup a packet's inner header. If the destination is the same as the ingress interface, the packet will be directed as normal. All future packets in the session are treated as vwire; as if overlay routing was not enabled. If the packet is going to an outbound destination, the firewall decapsulates the packet and forwards the packet to the IGW or NAT gateway. When the packet returns, the firewall reapplies the encapsulation.

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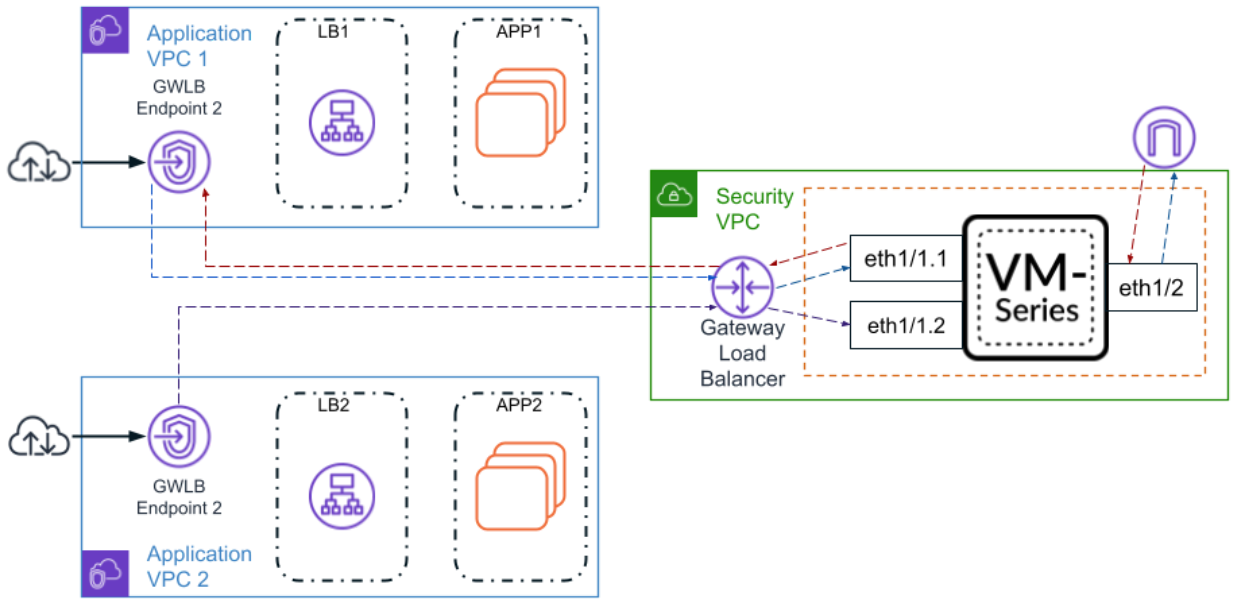

There is no SNAT in AWS.

XFF

6.9.1. AWS Inbound Design

For INBOUND traffic designs, consult the [Inbound Traffic Architecture]() document.

7. Deployment Phase

It may be easiest to deploy to the customer site using the cloud shell for the particular Public Cloud provider.

Note that not all cloud deployments are performed with Terraform. There can be manual or other types of deployments.

7.1. Design and Deployment Types

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Following are the internal links to the Public Cloud Reference Architecture for AWS, Azure, and GCP that covers different design and deployment types.. Below reference Architectures further breaks down into Design and Deployment Guides for each of the Cloud Providers.

Provider	**Reference Architecture**
AWS	Securing Applications with Cloud NGFW for AWS Reference Architecture
Azure	Securing Applications in Azure Reference Architecture
GCP	Securing Applications in Google Cloud Reference Architecture
Containers	Containers Reference Architecture
Azure Virtual WAN	Securing Microsoft Azure Virtual WAN

7.2. Panorama Preparation

The customer should decide on a strategy for routing management traffic and implement these tunnels and routing as needed before the deployment. This is required to manage the firewalls from Panorama. In other words, you probably do not want the management traffic to be routed over the inspection path, since that does not exist before the deployment.

If this is an existing Panorama, collect the management IP address and subnet CIDR. Record the information in the TRD.
You can use the Panorama CLI to generate VM Series auth key.

NOTE: use the max lifetime of 8760. The lifetime of the key can vary between 1 hour and 8760 hours (1 year). After the specified time, the key expires. Panorama does not register VM-Series firewalls without a valid auth key in the connection request.

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Collect the names of Device Group(s), Template(s) and Template Stack(s) the customer would like to use. A Device Group Name and Template Stack name can be included in the Terraform deployment code.

7.3. Terraform Preparation

The preferred method for deploying Terraform in the customer environment is to pass them a compressed file containing all the Terraform code. The customer can then upload this file to the cloudshell of the Public Cloud provider, where it can be unpacked.

The consultant should review the Terraform code files, especially the terraform.tfvars and variables.tf files to make sure everything is correct before the deployment.

- Panorama IP
- Management IP
- SSH keys
- CIDR blocks

The customer should provide a “public” SSH key that can be included in the terraform.tfvars file. This key will be used to log in to each new firewall via CLI session immediately after the firewall is provisioned. The exception is Azure, which will set a random password on the new instance. Azure also uses “panadmin” as the username, rather than “admin” which is used in all other public cloud providers.

Once the Terraform code has been completed by the Poland Scale team, it is recommended that the PSC set up a meeting with the developer. This gives the PSC a chance to ask questions, note any discrepancies, and understand what was done as far as testing. If everything looks good, the PSC can take delivery of the code and test it out in the lab before scheduling a deployment session with the customer. If you need help with reviewing or testing Terraform code, you can reach out to Franklin Diaz for assistance.

The PSC is strongly encouraged to test the customer traffic flows in our Public Cloud labs to the extent possible.

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7.4. Terraform Execution

Preparing to execute Terraform in the lab or in a customer site means reviewing all the variables and tfvars file to make sure they are correct. Do this review with the customer if possible, since it is common to find small issues with firewall naming, CIDR blocks, and so on.

Terraform execution is a three part process that includes init, plan, and apply.

When you run “terraform init” the “backend” is prepared. This includes the storage location for the modules that are downloaded and maintaining a snapshot of the current state.

Some consider running “terraform plan” an optional step but it can be helpful in finding and identifying issues.

Finally, the “terraform apply” command will provision the infrastructure as you have it in your local files.

7.5. Initial Firewall Configuration

Once a new Panorama or NGFW is deployed, you must set the Admin Password. It takes about 10 minutes for the new NGFW/Panorama to come up for the first time. Use the relevant Cloud Provider for the steps. Link below will take you to the AWS steps.

NOTE*: In Public Cloud deployments, default username password (admin/admin) will not work. VM Series PanOS images use the key pair provided during the instance deployment. Once MUST setup the initial admin password (step7) afterwards they are logged into via CLI or defined in terraform.tfvars prior to the deployment. Follow the steps in the link below in “Install Panorama on AWS”.*

Install Panorama on AWS
Install Panorama on Azure
Install Panorama on GCP

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There is an exception to this workflow, and that is with Azure. For Azure deployments, the username is “panadmin” and the password is typically set in the terraform.tfvars file. The password is often displayed in the Terraform output, concealed as a “sensitive value”.

7.6. Configure DNS and NTP

Customers may choose to provide DNS and NTP servers that they want to use in their Public Cloud Panorama and VM series. To prepare for licensing and connection to Panorama, these values should be added in “Device -> Setup -> Services” menu.

Public DNS servers can be used temporarily. Advise the customer that these values can be overwritten once the Panorama is connected and managing the virtual firewalls.

- 1.1.1.1
- 8.8.8.8
- 8.8.4.4

Public NTP servers can be used temporarily if needed. Advise the customer that these values can be overwritten once the Panorama is connected and managing the virtual firewalls.

- 0.pool.ntp.org
- 1.pool.ntp.org
- 2.pool.ntp.org
- 3.pool.ntp.org

7.7. Register the Firewalls in the Support Portal

The new FLEX licensing paradigm is the preferred way to license firewalls in PANOS release 10.2 and beyond.

Prior to the internal kickoff for the engagement, the Sales and Account teams will work with the customer to understand their requirements and traffic flows. This

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discussion should result in them guiding the customer towards a certain shape and number of NGFW to suit their needs. There is a calculator tool for the number of FLEX credits needed for the number of PANOS devices and features the customer wants.

Some customers may have had Auth Codes mailed to them from Palo Alto. The account team is responsible for assisting the customer with obtaining auth codes for their PANOS devices. If the customer is unable to locate their auth codes, ask them to work with the account team to resolve the issue.

PSC should work with the customer and the Palo Alto Account Team to understand which of these types of licensing the customer uses. To be very clear, you only need to follow one of the following sections, either FLEX or Auth Codes.

7.7.1. FLEX Licensing

Using the deployment profile you created in section 6 with the customer, register the firewalls in the support portal.

Once this step is completed, you should see a serial number on each firewall. Now you are ready to on-board the firewalls to Panorama.

7.7.2. Auth Codes

On each firewall, select “Device -> Licenses -> Retrieve License keys from license server”. This may take some time, and you may need to “f5 refresh” the browser window, and log in to the firewall again.

Once this step is completed, you should see a serial number on each firewall. Now you are ready to on-board the firewalls to Panorama.

7.7.3. Fetch Device Certificate

The device certificate can be fetched from the Customer Support Portal and applied to each PANOS device that you deploy in the customer environment, including Panorama and NGFW.

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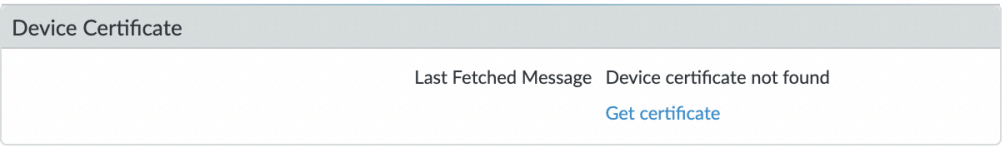

7.8. Connect to Panorama

Ensure the Panorama has the correct plug in installed to support the cloud provider the NGFW was deployed to. The version of the plug-in you use should match the PAN-OS running on Panorama. The Panorama is always on a higher release version or point release than the firewalls.

- Use the plug in compatibility matrix to verify you have the proper minimum plug-in versions installed on Panorama.
- You may choose to install plug-ins for more than one Public Cloud provider on a Panorama.
- You should install the plug in on Panorama HA peer as well to maintain HA connection.

On Panorama, under “Panorama -> Managed Devices -> Summary” Select “Add” at the bottom and paste in the serial number for each firewall, one per line. Leave the “associate devices” box checked. Click the button that says “copy auth key to clipboard”. Commit this change on the active Panorama. You should see the new firewalls listed in the Summary of the Managed Devices, perhaps in the “No Device Group Assigned” section of the page.

On each firewall, under “Device -> Setup -> Management -> Panorama Settings” add in the IP address(es) for Panorama. Paste in the key you generated previously. Commit this change on each firewall. Refresh the dashboard on each firewall. After about one minute, you should see “panorama” listed several times under the “Logged In Admins” section of the dashboard. If not, there may be an issue with the routing of the management traffic in the customer network.

The new firewalls should now be listed as “connected” in the Summary of the Managed Devices, Recommend that the customer perform a backup of the configuration changes at this point.

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7.9. Templates

Template Stacks are created on the Panorama from the “Panorama -> Templates” page. A Template Stack is used to group two or more Templates, which are created on the same page in Panorama. The “Template” section of the Panorama interface does not appear until you create the first template or stack.

The customer should also create one or more Templates, which can be added to the stack.

Edit the Template stack to include the recently onboarded firewalls and templates. Commit the change to Panorama. Once the commit is completed, push the change to the new firewalls. It is recommended that you edit the push to only include the relevant changes, especially if all users are logging in as “admin”. At this point, you should see the “template” column of the summary under managed devices “go green”.

7.10. Device Groups

The “Device Groups” section of the Panorama interface does not appear until you create the first device group.

Commit the change to Panorama. Once the commit is completed, push the change to the new firewalls. It is recommended that you edit the push to only include the relevant changes, especially if all users are logging in as “admin”. At this point, you should see the “shared policy” column of the summary under managed devices “go green”.

7.11. Create the Interfaces

The management interface e0/0 is not listed with the data interfaces of the virtual device.

AWS subinterfaces

7.12. Configure LB Health Checks

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7.12.1. AWS Health Check

Create a Interface Management Profile using the following steps:

- Select **Network** -> **Network Profiles** -> **Interface Mgmt**
- Click **Add** in the button left
- Be sure the **HTTP** checkbox is checked.
- Optionally, you can check the **PING** box.
- You can choose to set "Permitted IP Addresses" as well, but take care not to lock yourself out.

On the interface, add this management profile under the advanced tab. This should only be done on the "main" data interface, not the subinterfaces.

- Select the correct Template, then **Interfaces** -> **ethernet1/1** -> **Advanced** Tab.
- From the drop down, choose the management profile.

In the AWS portal:

- In AWS, navigate to **EC2** -> **load balancing** section.
- Under **Load Balancers** -> **target groups**, Click on the desired target group name, then click "register targets".
- Select the VM series from the list.
- Now check the "**include as pending below**" box.
- Finally, click the orange "**register targets**" box in the lower left corner.

The number of targets should update, along with their status as "healthy" or "unhealthy". You should also see the health checks in the "**Monitor**" tab of Panorama under "**Traffic**".

7.12.2. Azure Health Check

The health check should use port 80, not port 22. The response to the health check looks like incomplete sessions in the traffic logs.

Create an Interface Management Profile using the following steps:

- Select **Network** -> **Network Profiles** -> **Interface Mgmt**
- Click **Add** in the button left

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- Use the following configuration, updating the name to match customer preference:
    - Name: my-mp
    - Administrative Management Services: **SSH**
    - Permitted IP Addresses: **168.63.129.16/32**

Assign the profile to the Trust interface:

- Select **Network** -> **Interfaces** ->select the link for **ethernet1/2**
- Select the **Advanced** tab
- Set the Management Profile to **my-mp**
- Click **OK**

Assign the profile to the Untrust interface:

- Select **Network** -> **Interfaces** ->select the link for **ethernet1/1**
- Select the **Advanced** tab
- Set the Management Profile to: my-mp
- Click **OK**

Create a Static Route for the Azure Load Balancer Health Probes. Next we need to tell the health probes to flow out of the Untrust interface due to our 0.0.0.0/0 rule.

1. **Select Network -> Virtual Router -> Untrust-VR**
2. **Select Static Routes -> IPv4 -> Add**
3. **Use the following configuration**
    1. **Name: AzureLBHealthProbe**
    2. **Destination: 168.63.129.16/32**
    3. **Interface: ethernet 1/1**
    4. **Next Hop: IP Address**
    5. **IP Address: Use the IP address of the default gateway of your subnet the Trust interface is deployed on**
        1. **Note: To find this, navigate to the Azure Portal (portal.azure.com) and select All Services -> Virtual Networks -> Your Virtual**

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Network -> Subnets and use the first IP address of your subnet the trust interface is on. For example, if the address range of my subnet is 10.5.15.0/24, I would use 10.5.15.1 as my IP address. If my subnet was 10.5.15.128/25, I would use 129 10.5.15.129 as my IP address

4. **Click OK**

Recommend that the customer perform a backup of the configuration changes at this point.

7.13. Configure Virtual Routers

7.13.1. AWS Virtual Routers

The static routes under the AWS VR are left blank when using the GWLB architecture.

7.14. Initial Policy Configuration

Log forwarding profile

A policy is needed to permit health check probes from the public cloud provider.

Azure, section 3.11 of the deployment guide.

7.15. Enable Telemetry

PSC should enable telemetry for each NGFW and Panorama that is deployed. This is needed for AIOps and other “hub” features. The hub is located at https://apps.paloaltonetworks.com/apps

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8. Configuration Phase

Allowing traffic to traverse cloud infrastructure on designated paths is a two-part process. Security Groups and network routes need to be configured together for successful traffic flows. Security policy is the gating mechanism for traffic entering and exiting the firewall for inspection. Getting all these working together in accordance with best practices is crucial for ensuring the security of customer networks.

8.1. Configuring Security Policy

The log forwarding profile and health check policy between the load balancer(s) and the firewalls should already be configured at this point. This work was completed in a previous section.

Traffic policies are evaluated on the VM series firewall in a specific order as described in the following documentation.

- Packet Flow Sequence in PAN-OS - Knowledge Base - Palo Alto Networks
- Detailed Life of a Packet (Internal only, cannot be shared outside of GCS)

The goal is to evaluate customer network traffic based on the type of flow it is associated with. These typically include the following:

- Inbound
- Outbound
- East/West

8.2. Configuring Cloud Security Groups

Security groups are typically associated with subnets, dictating the incoming and outgoing traffic to be allowed.

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8.3. Cloud Routing

Routing in the cloud can be thought of in terms of their originating traffic flow type.

- Inbound
- Outbound
- East/West
- On Premise

Follow the internal confluence site for different design and deployment options.

Public Cloud - Validated Design Cookbook

These traffic types are configured using a combination of the following set of NGFW elements.

- Virtual Routers
- Traffic Policy
- NAT (SNAT/DNAT) Configurations
- Cloud Route Tables
- Cloud Security Groups

In Public Cloud deployments our goal is to add new Security infrastructure and ingest these traffic flows as “spokes” that feed into a central Security VPC/VNet.

8.3.1. AWS

It is recommended to use a non-overlay deployment. The Panorama configuration is simpler, we don’t need to use variables in the templates, and there is less concern in terms of routing. The use of sub-interfaces is recommended. This gives the ability to see each flow separately.

This Spring repository shows best practice/typical configuration for AWS Deployment:

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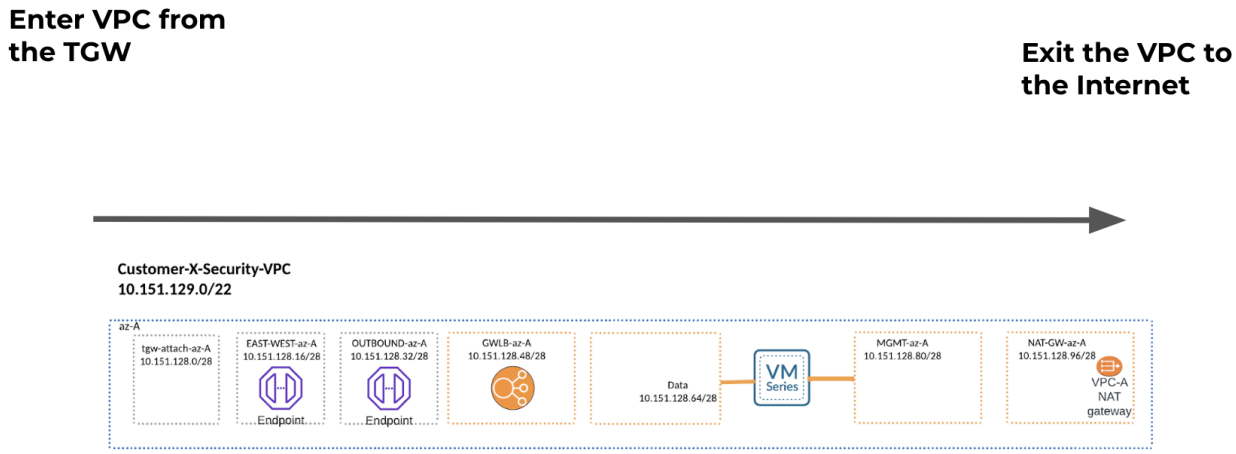
https://spring.paloaltonetworks.com/pso-prisma-automation/AWS-GWLB-validated-design

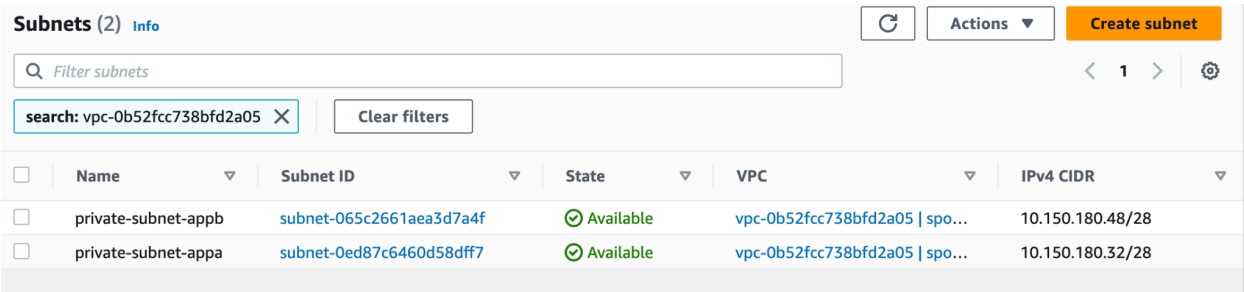
Refer to the AWS Traffic Flows for GWLB Deployments document for a more detailed description of this configuration.

8.3.1.1. AWS Outbound

Consider the following diagram from Sean Jolly's slides at the link mentioned above:



- Locate the application subnets in the application VPCs. For example:



- Verify that each subnet route table is configured to send traffic to the TGW:

Routes | Subnet associations | Edge associations | Route propagation | Tags

Routes (2)

Edit routes

Filter routes | Both | 1

Destination	Target	Status	Propagated
0.0.0.0/0	tgw-0630a51672d2cf1f6	Active	No
10.150.180.32/27	local	Active	No

- Get the TGW ID and navigate to Transit Gateway attachments. Filter on that TGW:

Transit gateway attachments (4) Info

Actions | Create transit gateway attachment

Filter transit gateway attachments | 1

search: tgw-0630a51672d2cf1f6 | Clear filters

Name	Transit gateway attachment ID	Transit gateway ID	Resource type	Resource ID	State
spoke-1	tgw-attach-08e5acfa030a3094d	tgw-0630a51672d2cf1f6	VPC	vpc-0311f2b7d52127729	Available
spoke-1	tgw-attach-0bd3a0fabf706d679	tgw-0630a51672d2cf1f6	VPC	vpc-01c06b21603ad856e	Available
spoke-2	tgw-attach-0f3a4a4c3cae47682	tgw-0630a51672d2cf1f6	VPC	vpc-0cd4c609916ce78a5	Available
my dog has fl;eas	tgw-attach-0fcab4657803298ad	tgw-0630a51672d2cf1f6	VPC	vpc-0b52fcc738bfd2a05	Available

- Navigate to TGW route tables and filter them using the same TGW ID. In this example we have two “from spoke” route tables since there is a separate inbound VPC. :

Transit gateway route tables (6) Info

Actions | Create transit gateway route table

Filter transit gateway route tables | 1

Transit gateway ID: tgw-0630a51672d2cf1f6 | Clear filters

Name	Transit gateway route table ID	Transit gateway ID	State	Default association route table
from_obew_sec_vpc	tgw-rtb-04fef58029842fee6	tgw-0630a51672d2cf1f6	Available	No
from_ib_sec_vpc	tgw-rtb-06ce4012682805119	tgw-0630a51672d2cf1f6	Available	No
from_ib_sec_vpc	tgw-rtb-078081894c3e45b99	tgw-0630a51672d2cf1f6	Available	No
from_spoke_vpc	tgw-rtb-0960054d212b5966b	tgw-0630a51672d2cf1f6	Available	No
from_spoke_vpc	tgw-rtb-0efe09a08cd9918c3	tgw-0630a51672d2cf1f6	Available	No
from_obew_sec_vpc	tgw-rtb-0f1f3fd23f1dffda5	tgw-0630a51672d2cf1f6	Available	No

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8.3.2. Azure

In Azure, routing is accomplished using User Defined Routes (UDRs). In Azure, a UDR is analogous to a route table. On Azure, UDRs are for traffic leaving a subnet only.

https://learn.microsoft.com/en-us/azure/virtual-network/virtual-networks-udr-overview

Subnets in Azure always have fixed/reserved addresses that cannot be used for firewalls, load balancers, etc. The smallest supported IPv4 subnet is /29, and the largest is /2 (using CIDR subnet definitions). IPv6 subnets must be exactly /64 in size. Azure reserves the first four and last IP address for a total of 5 IP addresses within each subnet.

For example, the IP address range of 192.168.1.0/24 has the following reserved addresses:

IP Address	Description
192.168.1.0	Network address
192.168.1.1	Reserved by Azure for the default gateway
192.168.1.2, 192.168.1.3	Reserved by Azure to map the Azure DNS IPs to the VNet space
192.168.1.255	Network broadcast address

Figure X - Azure subnet example reserved IP Addresses

- https://knowledgebase.paloaltonetworks.com/KCSArticleDetail?id=kA10g000000PLa3CAG
- https://docs.paloaltonetworks.com/vm-series/10-2/vm-series-deployment/set-up-the-vm-series-firewall-on-azure/about-the-vm-series-firewall-on-azure/azure-networking-and-vm-series

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8.4. Iron Skillet

Unless they are a new customer, they usually have baseline profiles. The benefit of running Iron Skillet is reports and profiles.

- IronSkillet Day One Configuration Template (Github Repo)
- IronSkillet Day One Configuration (Live community)

9. Documentation Phase

At this stage the deployment and configuration of the customer network has been completed. The consultant should review the TRD for completeness and finalize that document by reviewing for accuracy and completeness. The final project deliverable is the "As Built" document. This document is a description of the work performed for the customer, typically around 40 pages in length.

9.1. Presentation Resources

Resources to help with report writing and slide deck creation.

- 2023 Corporate Presentation Template
- Brand - Official Logos for use in Presenations
- Product Iconography
- Icon Library - Dark Mode

9.2. Cloud Documentation

Anton Coleman has a project going for this.

9.3. As Built Documents

The As Built document is the final deliverable in a Public Cloud Deployment. It is a report that details what work was accomplished during the engagement. The document is provided to the customer for their reference. On completion, the As Built should be saved in the Google Drive folder provided by the Project Manager.

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9.3.1. Collection of TSF files

Once the project is fully deployed, the Tech Support Files (TSF) are generated from the Panorama and all the newly deployed virtual firewalls. Configuration data is extracted from the XML configuration in each of these TSF files.

The consultant and/or the customer can use these steps to collect TSF files:

1. Go to Device > Support, or on Panorama, Panorama > Support.
2. Under Tech Support File, Click Generate Tech Support File.

   NOTE: The option will not be available if the user has Device Administrator privileges. You need SuperUser privileges to generate a Tech Support File from the Web UI.

3. Wait for the generation process to complete.

   NOTE: You can see the last time the Tech Support file was generated.

4. When the option appears, Click Download Tech Support File and save the file.
   Device > Support, showing you where to generate the Tech Support file from the WebGUI.

9.3.2. The SwiftDoc Tool

The SwiftDoc tool is used to generate an initial As Built template for engagements. The tool uses customer TSF files as input and creates a set of tables, links to documentation for the chosen PANOS release, and prompts for writing the final description of the project. The TSF files should be collected once the deployment and all related configuration work is completed. This will ensure that the most up to date state of the customer network is reflected.

https://theloop.paloaltonetworks.com/loop/swiftdoc

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9.3.3. Traffic Flow Diagrams

Traffic flow diagrams are included in the As Built document for each Public Cloud that was a part of the deployment activity.

Traffic flow diagram templates are detailed in the following table.

Public Cloud	Traffic Flow Template Link
Alicloud	
AWS	https://lucid.app/lucidchart/b184b420-8b6f-4276-b52f-09397fae0322/edit?invitationId=inv_8fbbb6d3-a859-4b0d-b7ea-3a6fcf0cfe36&page=LqzkG6wiqI_x#
Azure	
Google Cloud	
IBM	
Oracle	

Figure X - Traffic Flow Templates

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10. Links to Supporting Documentation

10.1. PS Content

- Google Drive folder containing Cloud Team documentation and artifacts.
- PS Employee Handbook describes many useful duties and responsibilities of the PS engineer.
- PSC Runbook describes some of the duties and expectations of a PS engineer.
- PS Licensing Updates
- Software NGFW Professional Services (VM / CN)
- PAN-OS Upgrade Checklist
- SecOps Optimization for NGFW OA&M
- PS Public Cloud Workshop - AWS Advanced
- Pub Cloud Traffic Testing

https://confluence.paloaltonetworks.local/display/InfoDump/Professional+Services+-+Software+NGFW
https://drive.google.com/drive/folders/1RphiSQ0yCPT2xlPvd7dLYXqZ-m8dgskn
https://drive.google.com/drive/folders/1iJ0BPBJ2dYiBI2Jtxm9GEupUSQqtwZ81
https://panw-gcs.slack.com/archives/CD46T1XRU
https://theloop.paloaltonetworks.com/loop/professional-services-software-ngfw
https://theloop.paloaltonetworks.com/loop/departments/global-customer-services/global-customer-services/global-customer-support-delivery/gcsd-product-specializations/welcome-to-the-gcs-virtualization-vm-series-page/vm-series-issue-tracker
https://theloop.paloaltonetworks.com/a/paloaltonetworks/loop/vm-series-news

https://confluence.paloaltonetworks.local/display/InfoDump/AWS+Resource+Monitoring+with+Panorama+Plugin

https://confluence.paloaltonetworks.local/display/InfoDump/Professional+Services+-+Software+NGFW

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11. List of Abbreviations

Abbreviation	Definition
AWS	Amazon Web Services
GCP	Google Cloud Platform
GP	Global Protect
GWLB	Gateway Load Balancer in AWS
HLD	High Level Design
NGFW	Next Generation Firewall
OCI	Oracle Cloud Infrastructure
PSC	Professional Services Consultant
QS	Quick Start
T & M	Time and Materials
TAC	Palo Alto support
TGW	Transit GateWay in AWS
TRD	Technical Requirements Document
TSF	Technical Support File
UDR	Azure Route Table
VNet	Virtual Network (private cloud) in Azure
VPC	Virtual Private Cloud in AWS or GCP

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12. Document Properties

This document is prepared for the sole use by Palo Alto Networks Professional Services Organization. Do not distribute outside of PAN-PSO.

12.1. Contributors

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Jan 23, 2023	0.1	Franklin Diaz, Sean Jolly, Sean Youngberg, Steve Steiner, Bilal Ashfaq	Initial Draft	Creating and populating the initial sections.
Nov 2, 2023	0.2	Franklin Diaz	Document Refresh	• Remove empty sections • Add link to traffic testing

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